MIL-M-10323A(MU)
16 October 1967
SUPERSEDING
MIL-M-10323(Ord)
25 May 1950

MILITARY SPECIFICATION

MOUNT, TELESCOPE: M48

1. SCOPE

1.1 This specification covers one type of telescope mount known as Mount, Telescope: M48.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U. S. Army, Frankford Arsenal

D6510712	Adapter Fixture Gage Assembly
D6523553	Azimuth Test Fixture
F6583165	Mount, Telescope, M48

PACKAGING DATA SHEET

MIL-P-14232/P6583165	Packaging of Mount, Telescope: M48

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(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The mount shall be manufactured in accordance with Drawing F6583165.

3.2 General specification.- The contractor shall be responsible for compliance with the following requirements of MIL-F-13926:

a. Order of precedence.

b. Dimensions and tolerances.

c. Inorganic protective surface finishes.

d. Part identification and marking.

3.3 Temperature conditions.- The requirements of this specification shall be met with the mount thermally stabilized at an ambient temperature of between +60 and +90° Fahrenheit (F) unless otherwise specified.

3.3.1 Extreme operating temperatures.- The mount shall meet the requirements of 3.7.3 while thermally stabilized at ambient temperatures of -40° and +150°F.

3.4 Vibration.- The mount shall be vibrated at a constant frequency of 30 cycles per second with a displacement of 1/8 inch for a period of 5 minutes plus or minus 15 seconds at room temperature plus 60° to 90° (F) subsequent to vibration all requirements of the specification shall be met.

3.5 Orientation.- The mount shall be set up as shown on the assembled viewing Drawing F6583165. The axis of the mounting spindle shall be vertical and the level centered within the etched circle on the cover glass.

3.6 Accuracy.

3.6.1 Azimuth reading.- Error of any azimuth reading shall not exceed 2.0 mils including backlash.

3.6.2 Backlash.- Backlash in the azimuth and orienting worm mechanism shall not exceed 1.0 mil in each mechanism.

3.6.3 Vertical displacement.- The axis of the mounting spindle shall not deviate from its vertical position by more than 1.5 mils when the mount is rotated to any position of azimuth rotation.

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3.6.4 Scale coincidence.- The zero graduation of the azimuth micrometer shall coincide with its index when any azimuth scale marking is in coincidence with the azimuth scale index.

3.7 Operability.

3.7.1 Azimuth knob.- The torque required to rotate the azimuth knob shall be within a minimum of 24 to a maximum of 36 inch-ounces. The maximum variation of the operational torque value shall not exceed plus or minus 4 inch-ounces over 6400 mils of mount travel.

3.7.2 Orienting knob.- The torque required to rotate the orienting knob shall be within a minimum of 32 to a maximum of 42 inch-ounces over 6400 mils of mount travel. The maximum variation of the operational torque valve shall not exceed plus or minus 6 inch-ounces over 6400 mils of mount travel.

3.7.3 High and low temperature operability.- With the mount subjected to the requirements of 3.3.1. The torque on the azimuth and orienting knobs shall not exceed 8 inch pounds at -40°F, at +150°F the maximum allowable torques as indicated in 3.7.1 and 3.7.2 shall not be exceeded.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article submission.- When a first article sample is required by the contract it shall be submitted to the Government for test and evaluation in accordance with the contract schedule. The sample shall consist of three (3) mounts. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, equipment, processes and including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply. After provisional acceptance at source, the initial production samples will be inspected and tested by the Government for all requirements and tests of the drawings and specifications. Government inspection and testing may be conducted at the contract facility or as otherwise specified by the contract. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

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4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination, and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II, and III herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II, and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 0.65% Defective		
101. Vibration	3.4	4.6.2;4.6.2.1
102. Azimuth reading	3.6.1	4.6.4.1
103. Backlash, azimuth worm mechanism	3.6.2	4.6.4.2
104. Backlash, orienting worm mechanism	3.6.2	4.6.4.3
105. Vertical displacement	3.6.3	4.6.4.4
106. Scale coincidence	3.6.4	4.6.4.5
107. Aximuth knob, operability	3.7.1	4.6.4.6
108. Orienting knob, operability	3.7.2	4.6.4.7
MINOR: NONE		

NOTE: The tests in Table I shall be conducted at a temperature between +60° and +90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special Sampling.

4.4.1 General.- One mount shall be selected at random as a special sample from each 100 produced. The sample shall meet the requirements and tests in Table II.

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TABLE II

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Fabrication	3.1	Applicable drawing - visual
302. General specification	3.2	MIL-F-13926 - visual

NOTE: The tests in Table II shall be conducted at a temperature between +60° and +90°F.

4.4.2 Environmental.- Three mounts shall be selected at random as a sample from each 50 produced, or from each month's production, whichever occurs first. The sample shall meet the requirements and tests in Table III and shall then meet the requirements and tests in Table I.

TABLE III

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303. Operating temperatures (-40° and + 150°F)	3.3.1	4.6.1
304. Torque, azimuth knob	3.7.3	4.6.4.8
305. Torque, orienting knob	3.7.3	4.6.4.8

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. - Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test facilities" requirements of MIL-F-13926.

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Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment to test performance.- The special testing equipment (STE) shall consist of an azimuth fixture, adapter fixture, perpendicularity check gage, viewing telescope and a test target.

(a) An azimuth testing fixture conforming to requirements of Drawing D6523553 employed to perform the azimuth tests, may be utilized.

(b) An adapter fixture, conforming to requirements of Drawing D6510712, employed for the purpose of mounting and holding the mount on the azimuth testing fixture, may be utilized.

(c) A perpendicularity check gage, conforming to requirements of Drawing D6510712, is used with a commercial bench level having a sensitivity of 5 seconds, in setting the axis of the mounting spindle vertically for tests. The gage shall have a "V" groove into which an auxiliary telescope fits to provide an optical line of sight which is an aid to perform the aximuth accuracy and vertical displacement tests.

(d) A test target consisting of a horizontal and vertical centerline with mil blocks extending in both directions in elevation and azimuth. The size of one mil, in inches, is determined by the distance at which the target is placed, from the objective of the viewing telescope, and is equal to the distance divided by a thousand. The mil blocks are utilized to measure backlash, azimuth error and vertical displacement.

(e) A viewing telescope of at least 3 power with a suitable cross line reticle will be used. The reticle shall be centered with respect to the outside diameter within 10 seconds. Provisions shall be made to permit plumbing the reticle within 10 seconds, and for maintaining this accuracy with respect to the perpendicularity gage and with the target.

4.6 Test methods and procedures.

4.6.1 Extreme operating temperatures.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926, and the conditions of 3.3.1. The mount shall be exposed to each temperature extreme for a minimum of 5 hours. The temperatures shall be changed gradually to avoid thermal shock. At the end of each 5 hour period, and while still at the temperature extreme, the mount shall be visually examined for damage and then tested as specified in 4.6.4.8.

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4.6.2 Vibration.- The mount shall be subjected to the vibration test utilizing a standard type of vibration machine capable of meeting the conditions of 3.4. The accuracy of the applicable test equipment and test equipment measurements shall be in conformance with the "Test Facilities" requirements of MIL-F-13926. The mount shall be secured to the vibrating table simulating its intended use by means of an adapter or holding media. The mount shall be vibrated in a vertical plane with the motion and frequency applied for the time indicated in 3.4. At the conclusion of this vibration, the mount shall be subjected to the examination in 4.6.2.1. The mount shall then be subjected to the tests and procedures outlined herein to determine compliance with the requirements of 3.4.

4.6.2.1 Examination, post vibration.- The mount shall be examined for physical defects resulting from poor workmanship and material. A visual examination shall be made for missing and damaged parts. A tactile examination shall be made to determine improper functioning, looseness or binding of the azimuth or orienting knobs. Anyone of the aforementioned possible defects shall be cause to consider a mount defective. Mounts which have passed the above examinations shall be subjected to the tests and procedures specified in 4.6.3 to 4.6.4.7 inclusive.

4.6.3 Orientation.- Pre-test adjustments of the special testing equipment depicted in 4.5.2.3. By the means provided, centralize and fasten the adapter fixture to the azimuth testing fixture. Adjust fixture to rotate in a horizontal plane. This is accomplished by means of a sensitive bench level and the rotation of the circular table of the azimuth fixture. The circular plate while rotated 360° must remain level, and the bench level bubble must reamin stationary. Attach the mount to the adapter fixture. Place the perpendicularity gage on the spindle of the mount and place the bench level on the perpendicularity gage. Adjust the mount until the spindle is vertical as indicated by the bench level in two positions 90° apart. When spindle is made vertical, firmly lock the mount to the adapter fixture, by means of its wing nut. With this accomplished, place the viewing telescope in the "V" provided in the perpendicularity gage. Set target at the distance for which the mil block size was calculated. Adjust the target until the center of the target is in line with the intersection of the telescope reticle. Plumb target by means of a plumb line without disturbing the alinement. With the spindle vertical as indicated by the perpendicularity gage and bench level as described above, the circular level bubble should be centered with respect to its etched circle.

4.6.4 Accuracy.

4.6.4.1 Azimuth reading.- The mount shall be inspected for compliance with 3.6.1 by taking at least eight random settings from the position established in 4.6.3, throughout the 6400 mils of travel in azimuth. With the scales of the mount and azimuth testing fixture indicating zero, offset the azimuth testing fixture an amount equal to a whole mil and then, by means of the azimuth knob of the mount, return the line of sight back into coincidence with the vertical centerline of the target while viewing

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through the auxiliary telescope. Azimuth error can be read on the azimuth knob micrometer scale, and shall be noted. Continue to offset the testing fixture an amount equal to a whole mil at a time, and recentering the auxiliary telescope's vertical reticle line with the vertical center line of the target by means of the azimuth knob. At each setting, the reading shall be noted. Error in any reading shall not exceed the limit specified in 3.6.1.

4.6.4.2 Backlash azimuth mechanism.- With the mount set up in the special testing equipment as in above test, ascertain that the viewing telescope's verticle reticle line is in coincidence with the vertical center line of the target. Backlash shall be checked at positions at which accuracy readings are taken. Rotate the azimuth knob about 800 mils away from the vertical target line and then rotate the knob in the opposite direction until the vertical reticle line of the telescope is again in coincidence with the vertical target line. Note the reading of the azimuth micrometer scale. Continue rotating the mount in the same direction about 300 mils and then reverse the direction of rotation until the vertical reticle line is in coincidence with the same vertical target line without overtravel. Again note the reading of the micrometer scale. The difference of the two readings is the total amount of backlash. Backlash for the mechanism shall not exceed the tolerance specified in 3.6.2.

4.6.4.3 Backlash orienting mechanism.- The test for backlash in the orienting mechanism is performed in a similar manner as in 4.6.4.2, except that it is necessary to scribe a line on the rim of the orienting knob and also improvise an index by scribing a line on a block placed adjacent to the rim of the scribed knob. Rotate the orienting worm in one direction until the scribed lines are in coincidence. Now rotate the azimuth testing fixture until the line of sight of the viewing telescope coincides with the vertical reticle line of the target. Continue to rotate the orienting worm one more revolution in the same direction. Reverse the direction of the worm knob one revolution until the scribed lines are again in coincidence without overtravel. Sight through the telescope and determine the extent of backlash as measured by means of the mil blocks on the target. The extent of backlash shall not exceed the tolerance specified in 3.6.2. Backlash shall be checked at positions at which accuracy readings are taken.

4.6.4.4 Vertical displacement.- The test is performed with the scales of the mount and the azimuth fixture set at zero, and the line of sight of the viewing telescope in coincidence with the intersection of the horizontal and vertical centerlines of the target. The azimuth fixture shall be deflected to a certain reading in mils past its starting point, away from the vertical target line. The azimuth knob of the mount shall be rotated in the opposite direction until the mount has traveled an equal distance as indicated by the azimuth micrometer scale. Observe the line of sight of the telescope in relation to the target. Indication of deviation as measured on the target shall be noted, and shall not exceed the limit specified in 3.7.3. Unless otherwise specified, the procedure shall be repeated a minimum of 8 readings taken at approximately equidistant points throughout the 6400 mils of mount travel. Deviation from

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the target line at any reading from any point shall not exceed the limit specified in 3.7.3.

4.6.4.5 Scale coincidence.- The inspection shall be accomplished from the position established in 4.6.3, with the scales of the mount and azimuth testing fixture indicating zero. Azimuth scale setting is a visual test that shall be made to verify the zero reading of the micrometer with the requirements of 3.6.4.

4.6.4.6 Operability azimuth knob.- A torque wrench type tester that will permit reading in both clockwise and counterclockwise directions measuring in inch-ounces, the accuracy of which shall be ± 0.25 inch-ounces throughout the entire range of movement shall be utilized. The torque wrench shall be equipped with the adapters to accommodate testing of the respective knobs. With the torque wrench and adapter attached to the azimuth knob, the measured torque required to operate the knob, at temperature condition of 3.3, shall be noted and shall be within the limits specified in 3.7.1. Torque, over the 6400 mils of mount travel shall be accomplished by means of the torque wrench rotated in a clockwise and then in a counterclockwise direction. The measured torque required to rotate the azimuth knob in any position throughout the 6400 mils of mount travel shall be noted. Any variation between these readings and the operational reading taken above shall not be less or exceed the tolerance specified in 3.7.1.

4.6.4.7 Operability orienting knob.- With the mount stabilized at the temperature condition of 3.3, the torque required to rotate the orienting knob shall be checked with the appropriate adapter and the torque wrench type tester conforming to the accuracy specified in 3.7.2. The measured torque required to rotate the knob through 6400 mils of mount travel, in the clockwise and counterclockwise directions, shall be noted. Any variation between these readings and the operational reading shall not be less or exceed the tolerance specified in 3.7.2.

4.6.4.8 High and low temperature operability.- The knobs shall conform to the operability requirements for each temperature extreme. The torque values for each knob obtained at each temperature extreme shall be recorded for comparison for those values obtained in 4.6.4.6 and 4.6.4.7. The maximum torque recorded during the performance testing of each respective knob, shall not exceed the value specified for the temperatures indicated in 3.7.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P6583165. The level of protection shall be as specified in the procurement document.

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6. NOTES

6.1 Intended use.- The Mount Telescope: M48 is intended to mount and support the Telescope, Battery Commanders M65 on the Tripod M10 or Tripod M17. The mount establishes the verticularity of the Battery Commander's Telescope and provides azimuth movement of the telescope over 6400 mils with graduation to 1 mil.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Selection of applicable levels of preservation, packaging and packing.

c. Applicable stock number.

Custodian:

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Preparing activity:

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